



## U50SI1 Schematics Rev:C

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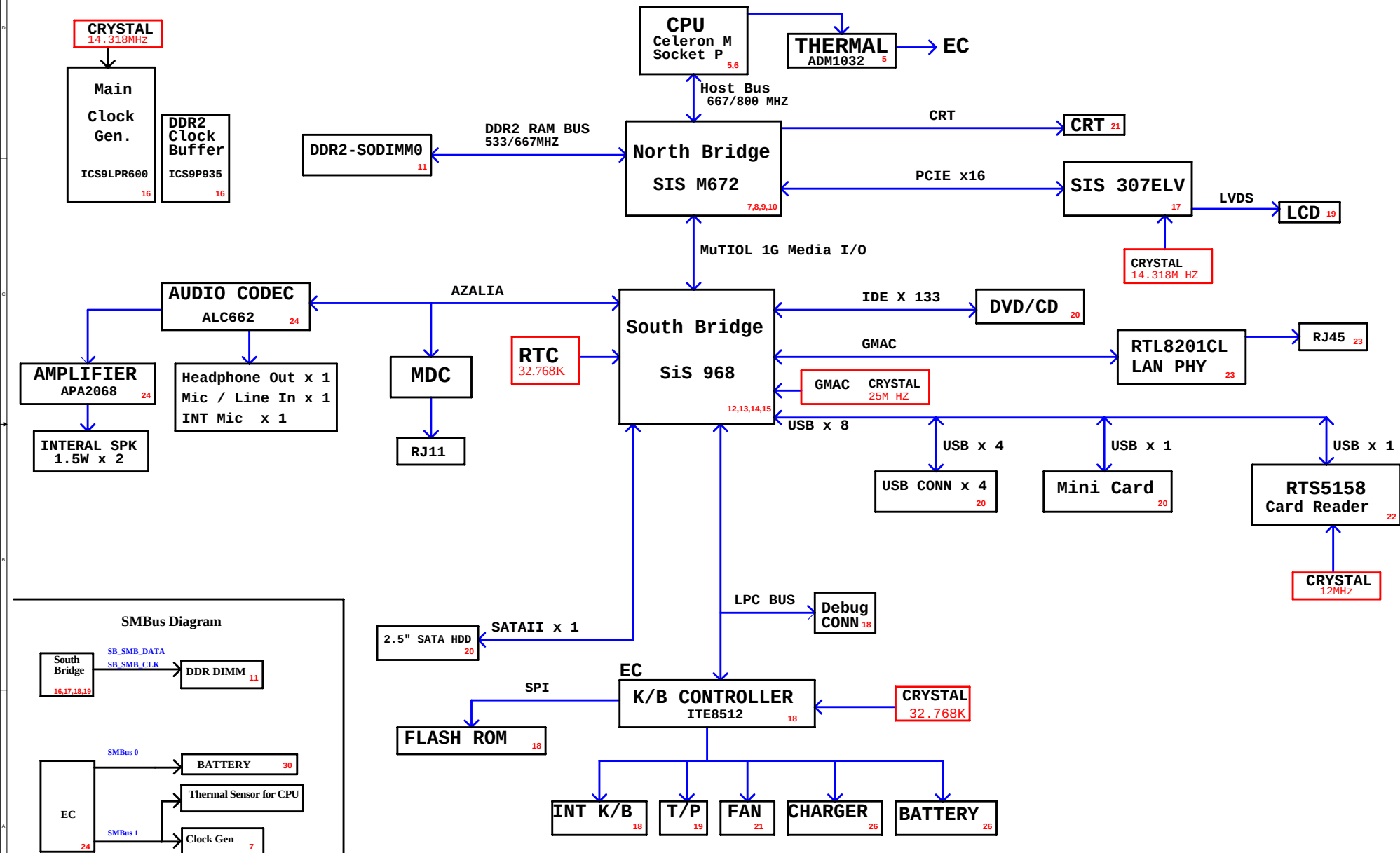
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### U50SI1 REV:C P/N LIST

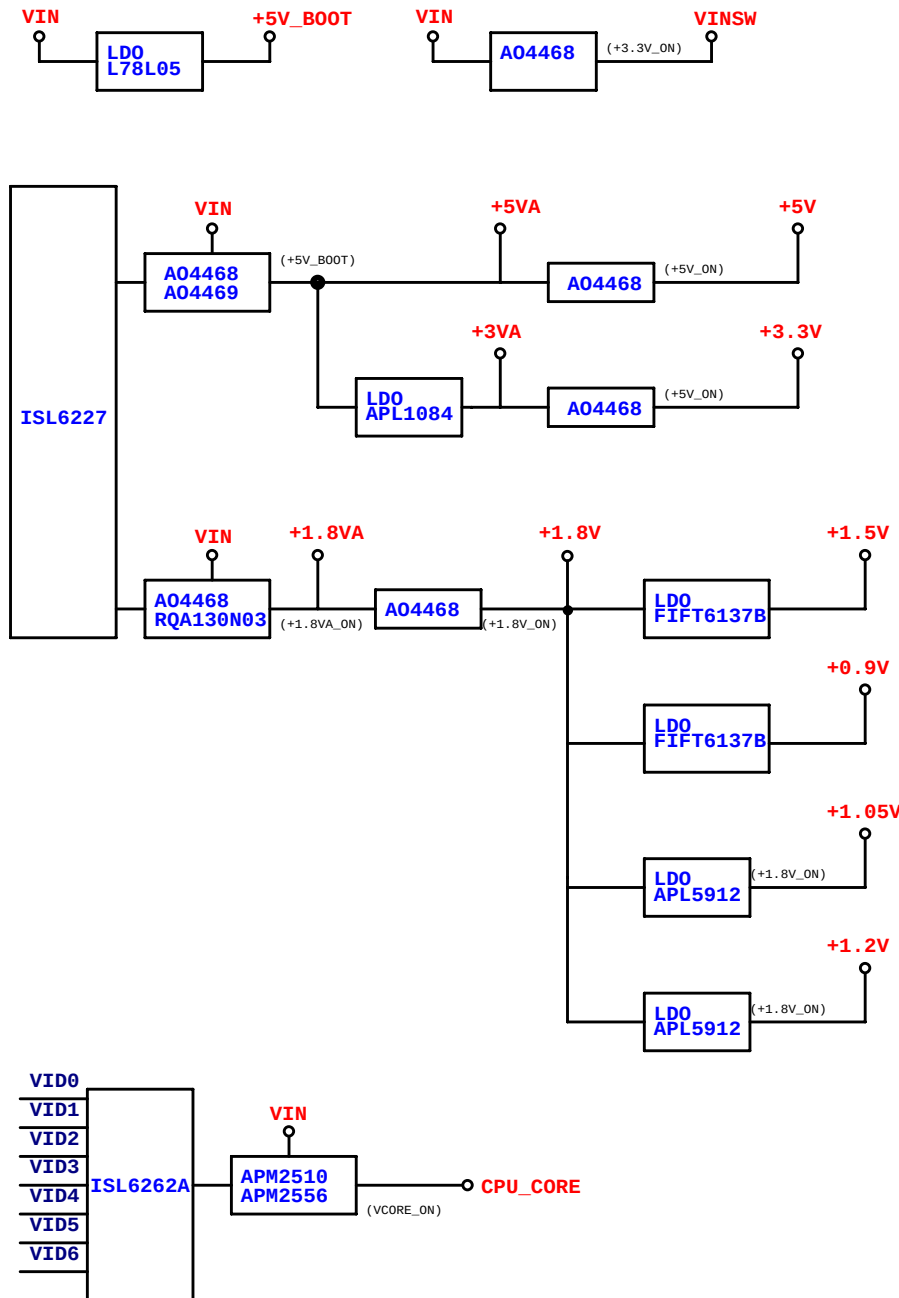
|  | PCB P/N | PCB ASSY P/N |
|--|---------|--------------|
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# U50SI1

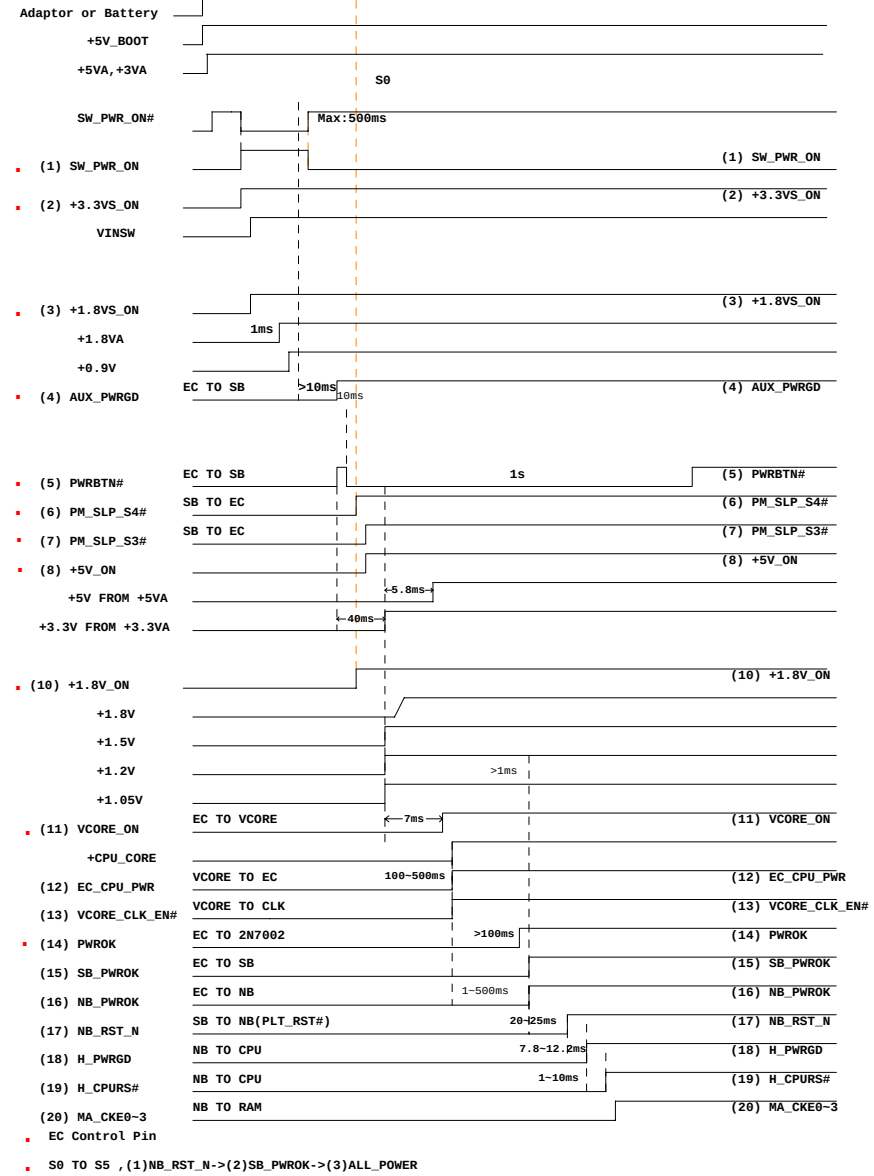
## SYSTEM BLOCK DIAGRAM



# POWER BLOCK DIAGRAM



# POWER Sequence



| SIS968<br>GPIO |               |
|----------------|---------------|
| GPI00          | NC            |
| GPI01          | NC            |
| GPI02          | PM_THROTTING# |
| GPI03          | EC_EXTSMI#    |
| GPI04          | PM_CLKRUN#    |
| GPI05          | NC            |
| GPI06          | NC            |
| GPI07          | NC            |
| GPI08          | NC            |
| GPI09          | NC            |
| GPI010         | SLP_S5#       |
| GPI011         | AGPSTOP_N     |
| GPI012         | DPSLP#        |
| GPI013         | SB DPRSLPVR   |
| GPI014         | NC            |
| GPI015         | SLP_S3#       |
| GPI016         | NC            |
| GPI017         | H_A20GATE     |
| GPI018         | H_RCIN#       |
| GPI019         | SB_SMB_CLK    |
| GPI020         | SB_SMB_DATA   |

| ITE8512E<br>GPIO |                    |
|------------------|--------------------|
| GPA0             | AUX_PWRGD          |
| GPA1             | DDR_V_SW#          |
| GPA2             | BTL_BEEP           |
| GPA3             | RFLED_ON           |
| GPA4             | SCROLL/3G_LED      |
| GPA5             | NUM_LED            |
| GPA6             | CAPS_LED           |
| GPA7             | PWRON_LED          |
| GPB0             | PM_SLP_S5#         |
| GPB1             | PM_SLP_S3#         |
| GPB2             | WEBCAM_ON          |
| GPB3             | BAT_SMBCLK         |
| GPB4             | BAT_SMBDAT         |
| GPB5             | H_A20GATE          |
| GPB6             | H_RCIN#            |
| GPB7             | BT_ON              |
| GPC0             | EC_VID5            |
| GPC1             | SMBCLK_EC          |
| GPC2             | SMBDAT_EC          |
| GPC3             | EC_VID2            |
| GPC4             | RF_SW_ON#          |
| GPC5             | EC_VID1            |
| GPC6             | INTERNET#          |
| GPC7             | SILENT#            |
| GPD0             | EC_PREST#          |
| GPD1             | PWRBTN#            |
| GPD2             | EC_LPCRST#         |
| GPD3             | EC_EXTSCI#         |
| GPD4             | EC_EXTSMI#         |
| GPD5             | H_PROCHOT#         |
| GPD6             | CHG_ON             |
| GPD7             | LCDSW              |
| GPE0             | EC_PWR_ON          |
| GPE1             | SET_V              |
| GPE2             | PWROK              |
| GPE3             | VCORE_ON           |
| GPE4             | LID#               |
| GPE5             | AC_IN/OUT#         |
| GPE6             | FAN_SPD# or RTCRST |
| GPE7             | AMP_MUTE#          |
| GPFO             | 3G_ON              |
| GPFI             | EC_BSEL1           |
| GPFF             | CHG_G_LED          |
| GPFF             | CHG_R_LED          |
| GPFF             | TP_CLK             |
| GPFF             | TP_DATA            |
| GPFF             | VGA_SMBCLK         |
| GPFF             | VGA_SMBDAT         |
| GPFO             | EC_VID3            |
| GPFI             | EC_WDOG_OK         |
| GPFG             | FLFRAME#           |
| GPFG             | NEW_CARD_PWR_ON#   |
| GPH0             | +1.8V_ON           |
| GPH1             | +1.8VS_ON          |
| GPH2             | SENBAT_V           |
| GPH3             | +3.3VS_ON          |
| GPH4             | +5V_ON             |
| GPH5             | VDD_CORE_ON        |
| GPH6             | EC_VID4            |

del VGA\_TEMP

| ITE8512E<br>GPIO |               |
|------------------|---------------|
| GPI0             | BATT_TEMP     |
| GPI1             | ADAPTOR_I     |
| GPI2             | BAT_V         |
| GPI3             | CPPE#         |
| GPI4             | BAT_I         |
| GPI5             | EC_CPU_PWR    |
| GPI6             | DDR2_TEMP     |
| GPI7             | ADAP_IN       |
| GPJ0             | EC_BRGHT      |
| GPJ1             | CHG_I         |
| GPJ2             | FAN_CTRL0     |
| GPJ3             | SILENT_LED    |
| GPJ4             | SMP1_EN#      |
| GPJ5             | PM_THROTTING# |

| CPU    |         |         |      |          |
|--------|---------|---------|------|----------|
| CPU    | CORE(V) | ICC(mA) | W    | TEMP(°C) |
| 2.0G   | 1.525   | 35.7    | 54.3 | 69       |
| 2.2G   | 1.525   | 37.5    | 57.1 | 70       |
| 2.26G  | 1.525   | 38.1    | 58.0 | 70       |
| 2.4G   | 1.525   | 39.3    | 59.8 | 71       |
| 2.5G   | 1.525   | 40      | 61.0 | 72       |
| 2.53G  | 1.525   | 40.4    | 61.5 | 72       |
| 2.6G   | 1.525   | 41.05   | 62.6 | 72       |
| 2.66G  | 1.525   | 43.35   | 66.1 | 74       |
| 2.8G   | 1.525   | 44.86   | 68.4 | 75       |
| 3.06G  | 1.525   | 55.9    | 85.2 | 81       |
| VCC    |         | ICC(mA) | W    | TEMP(°C) |
| +1.5V  | 120     | 0.18    |      |          |
| +1.05V | 2500    | 2.625   |      | 70       |

| 672MX  |         |       |          |
|--------|---------|-------|----------|
| VCC    | ICC(mA) | W     | TEMP(°C) |
| +1.2V  | 2303    | 2.76  |          |
| +1.8V  | 1215    | 2.18  | 70       |
| +1.05V | 80      | 0.084 |          |

| SIS968 |         |       |          |
|--------|---------|-------|----------|
| VCC    | ICC(mA) | W     | TEMP(°C) |
| +3.3V  | 86      | 0.283 |          |
| +1.8V  | 851     | 1.531 | 70       |
| +1.05V | 22      | 0.022 |          |

| 307LV |         |       |          |
|-------|---------|-------|----------|
| VCC   | ICC(mA) | W     | TEMP(°C) |
| +3.3V | 236     | 1.107 |          |
| +1.8V | 565     | 0.778 | 70       |

SMART POWER TABLE

| VID6 | VID5 | VID4 | VID3 | VID2 | VID1 | VID0 | VCORE  | +_mV   |
|------|------|------|------|------|------|------|--------|--------|
| 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1.5000 | -0mV   |
| 0    | 0    | 0    | 0    | 0    | 0    | 1    | 1.4875 | -2.5mV |
| 0    | 0    | 0    | 0    | 0    | 1    | 0    | 1.4750 | -5mV   |
| 0    | 0    | 0    | 0    | 1    | 0    | 0    | 1.4500 | -50mV  |
| 0    | 0    | 0    | 1    | 0    | 0    | 0    | 1.4000 | -100mV |
| 0    | 0    | 1    | 0    | 0    | 0    | 0    | 1.3000 | -200mV |
| 0    | 1    | 0    | 0    | 0    | 0    | 0    | 1.1000 | -400mV |
| 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0.7000 | -800mV |
| 0    | 0    | 1    | 1    | 0    | 1    | 1    | 1.1625 |        |
| 0    | 0    | 1    | 0    | 0    | 0    | 1    |        |        |
| 0    | 0    | 1    | 0    | 0    | 1    | 0    |        |        |
| 0    | 0    | 1    | 0    | 1    | 0    | 0    |        |        |
| 0    | 0    | 1    | 0    | 1    | 1    | 0    |        |        |
| 0    | 0    | 1    | 1    | 0    | 0    | 1    |        |        |
| 0    | 0    | 1    | 1    | 0    | 1    | 0    |        |        |

| CLOCK GENERATOR+BUFFER |         |      |          |
|------------------------|---------|------|----------|
| VCC                    | ICC(mA) | W    | TEMP(°C) |
| +3.3V                  | 400     | 1.32 |          |
| +1.8V                  | 300     | 0.54 | 70       |

| ITE8512E |         |      |          |
|----------|---------|------|----------|
| VCC      | ICC(mA) | W    | TEMP(°C) |
| +3.3V    | 200     | 0.66 |          |
| +3.3VA   | 500     | 1.65 | 70       |

| RTS5158 |         |      |          |
|---------|---------|------|----------|
| VCC     | ICC(mA) | W    | TEMP(°C) |
| +5V     | 76      | 0.38 | 85       |

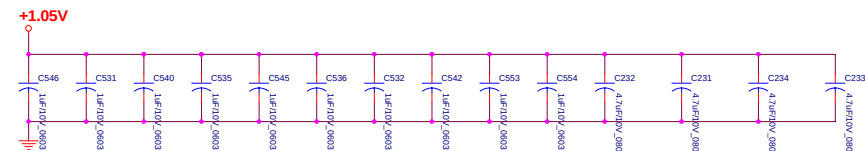
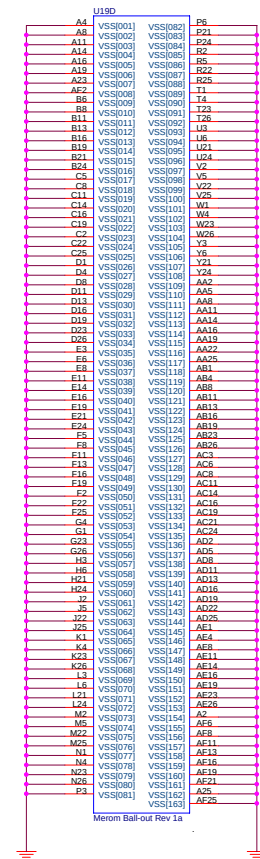
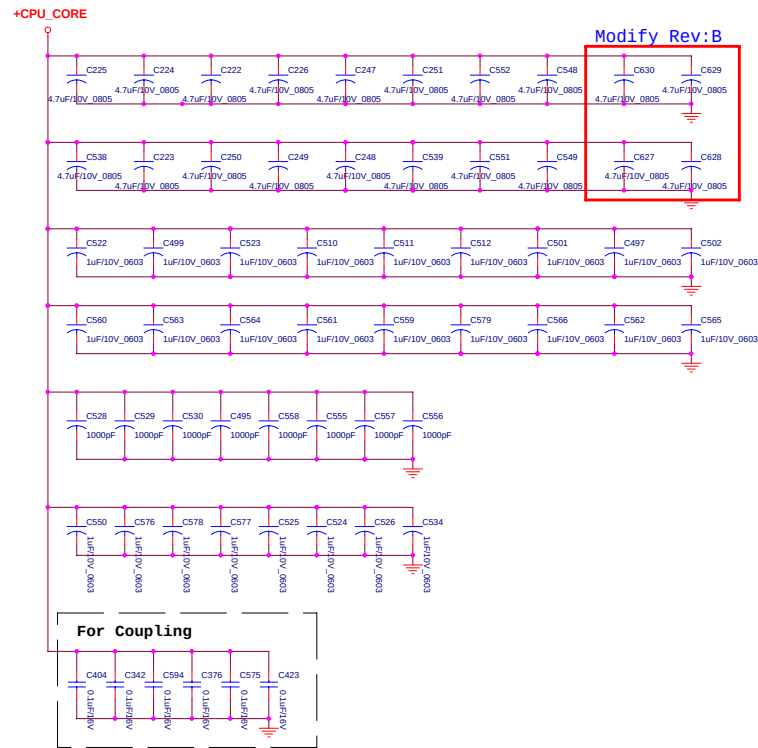
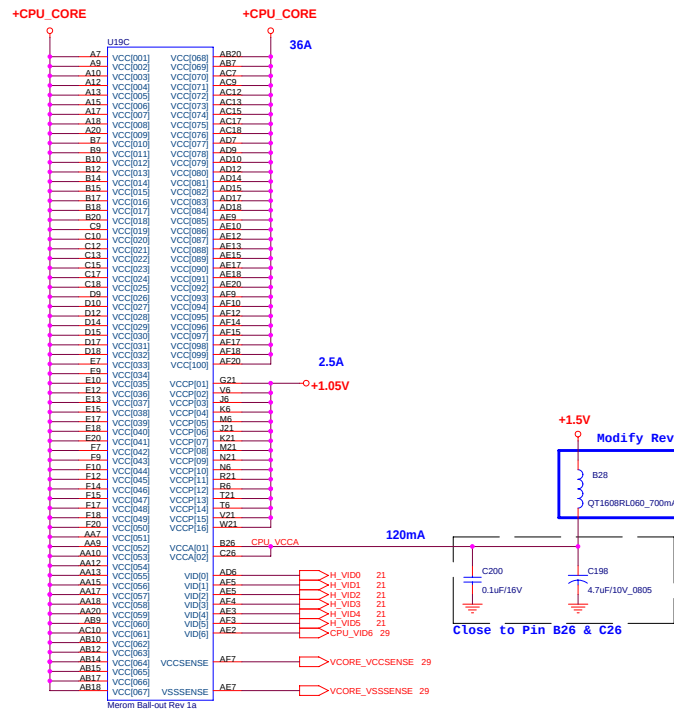
| RTL8201CL |         |       |          |
|-----------|---------|-------|----------|
| VCC       | ICC(mA) | W     | TEMP(°C) |
| +3.3V     | 20      | 0.396 | 85       |

| ALC662 |         |       |          |
|--------|---------|-------|----------|
| VCC    | ICC(mA) | W     | TEMP(°C) |
| +3.3V  | 23      | 0.075 |          |
| +5VA   | 38      | 0.19  | 70       |

| APA2068 |         |     |          |
|---------|---------|-----|----------|
| VCC     | ICC(mA) | W   | TEMP(°C) |
| 5V      | 20      | 0.1 | 85       |

| ADM1032 |       |        |          |
|---------|-------|--------|----------|
| VCC     | ICC   | W      | TEMP(°C) |
| +3.3V   | 170uA | 0.56mW | 150      |













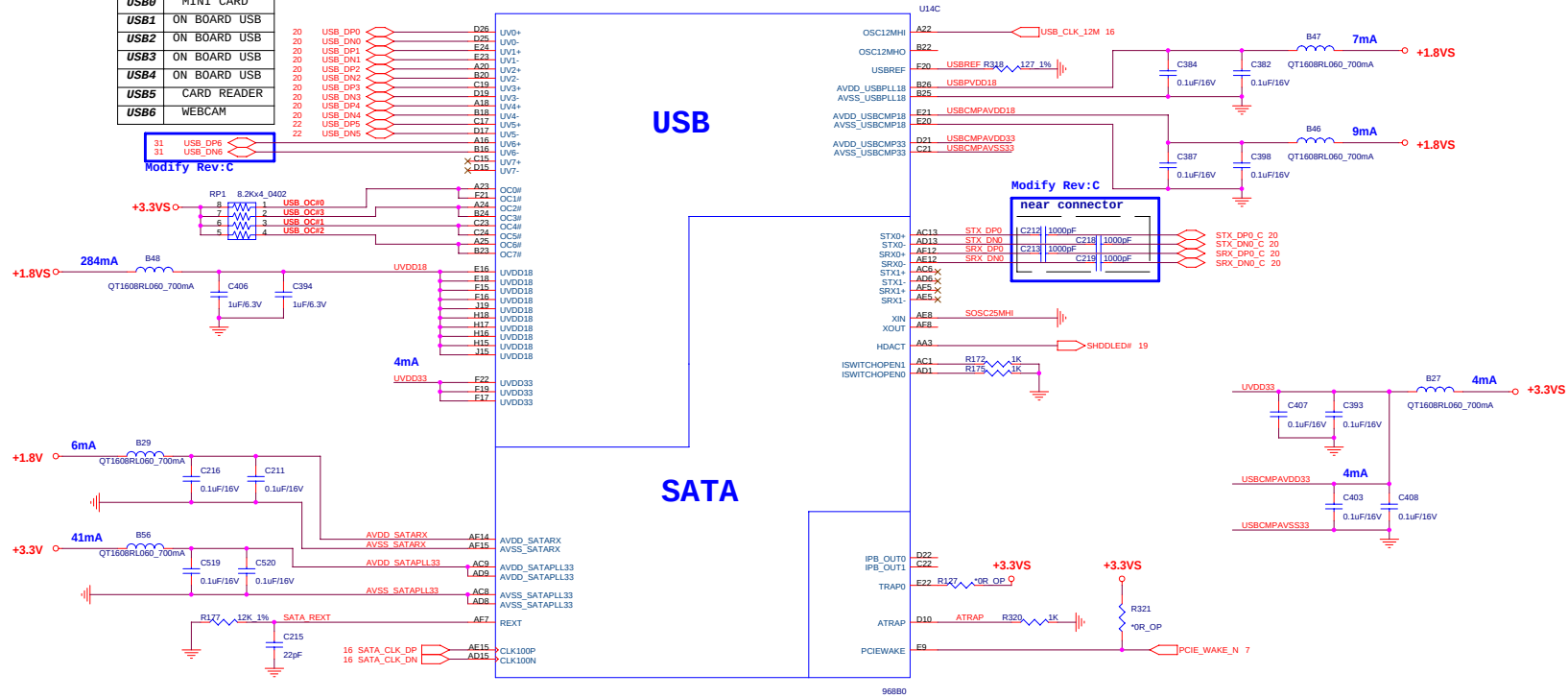




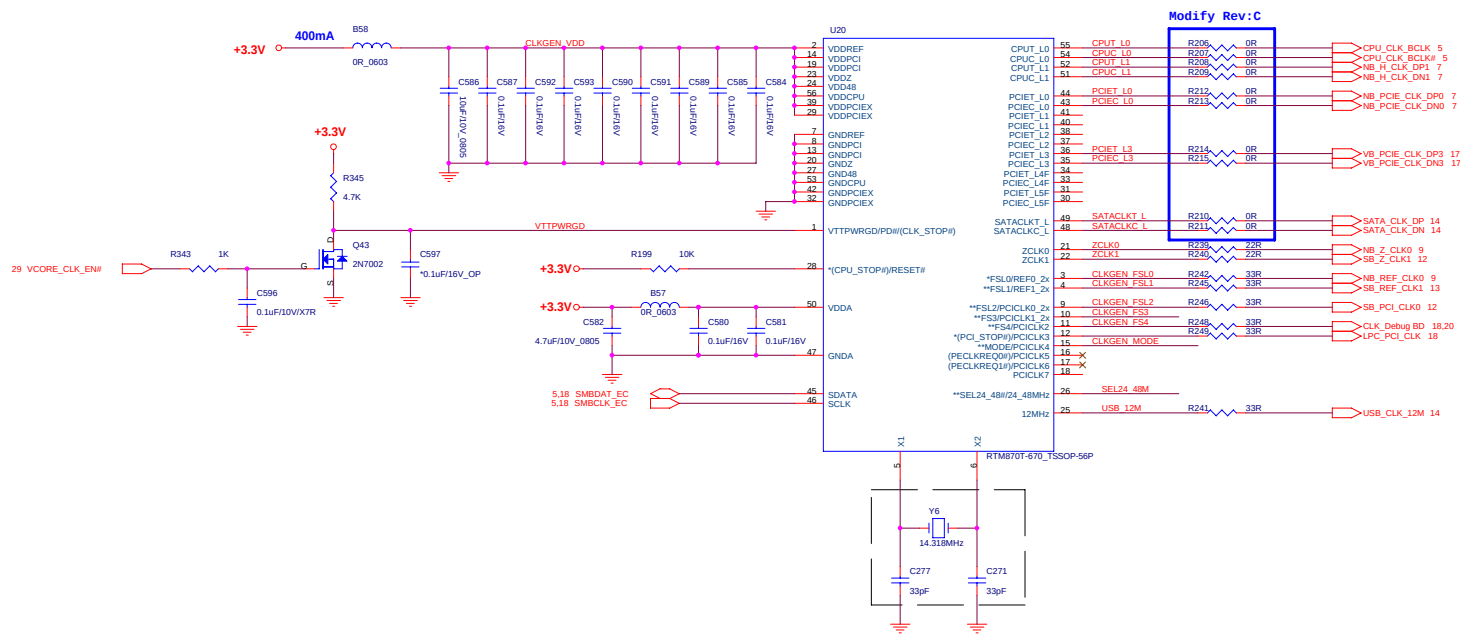




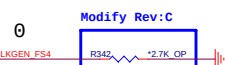
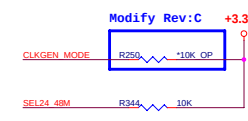
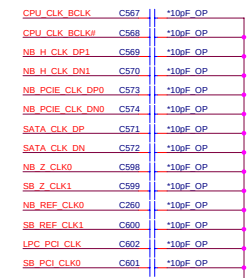
| U50SI0 |              |
|--------|--------------|
| USB0   | MINI CARD    |
| USB1   | ON BOARD USB |
| USB2   | ON BOARD USB |
| USB3   | ON BOARD USB |
| USB4   | ON BOARD USB |
| USB5   | CARD READER  |
| USB6   | WEBCAM       |



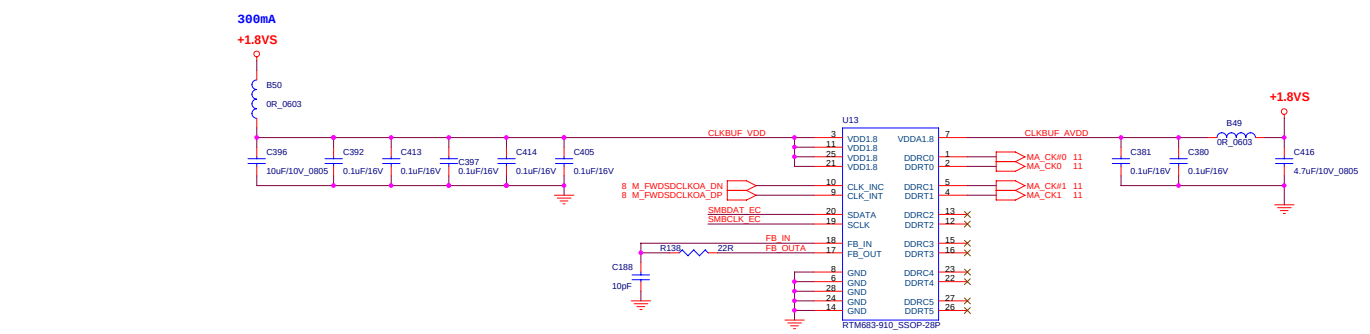




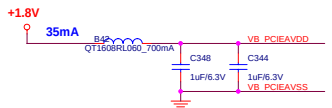
Please base on your design to choose the appropriate capacitor value.  
 $C_{210} + C_{X1} + C_{trace1} = C_1$   
 $C_{211} + C_{X2} + C_{trace2} = C_2$   
 $C_{load} (refer\ to\ the\ crystal\ datasheet) = (C_1 \cdot C_2) / (C_1 + C_2)$



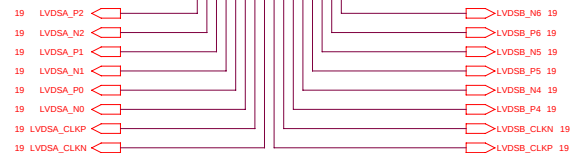
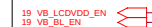
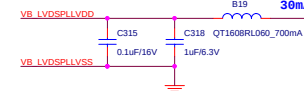
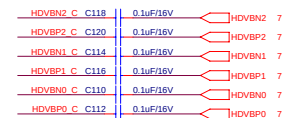
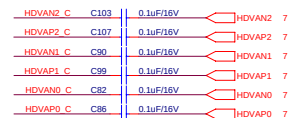
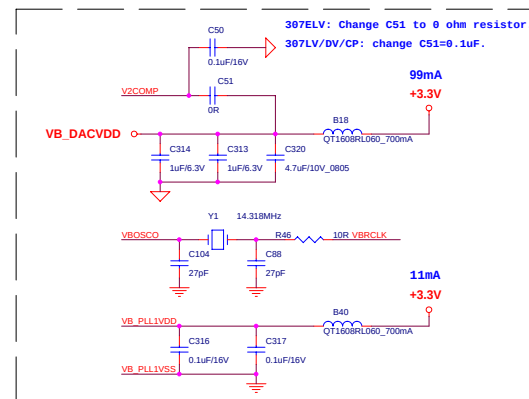
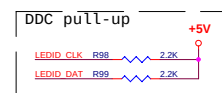
|        | BSEL4 | BSEL3 | BSEL2 | BSEL1 | BSEL0 | CPU_MHZ | ZCLK |
|--------|-------|-------|-------|-------|-------|---------|------|
| FSB533 | 0     | 0     | 0     | 0     | 1     | 133     | 133  |
| FSB667 | 0     | 0     | 0     | 1     | 1     | 166     | 133  |
| FSB800 | 0     | 0     | 0     | 1     | 0     | 200     | 133  |

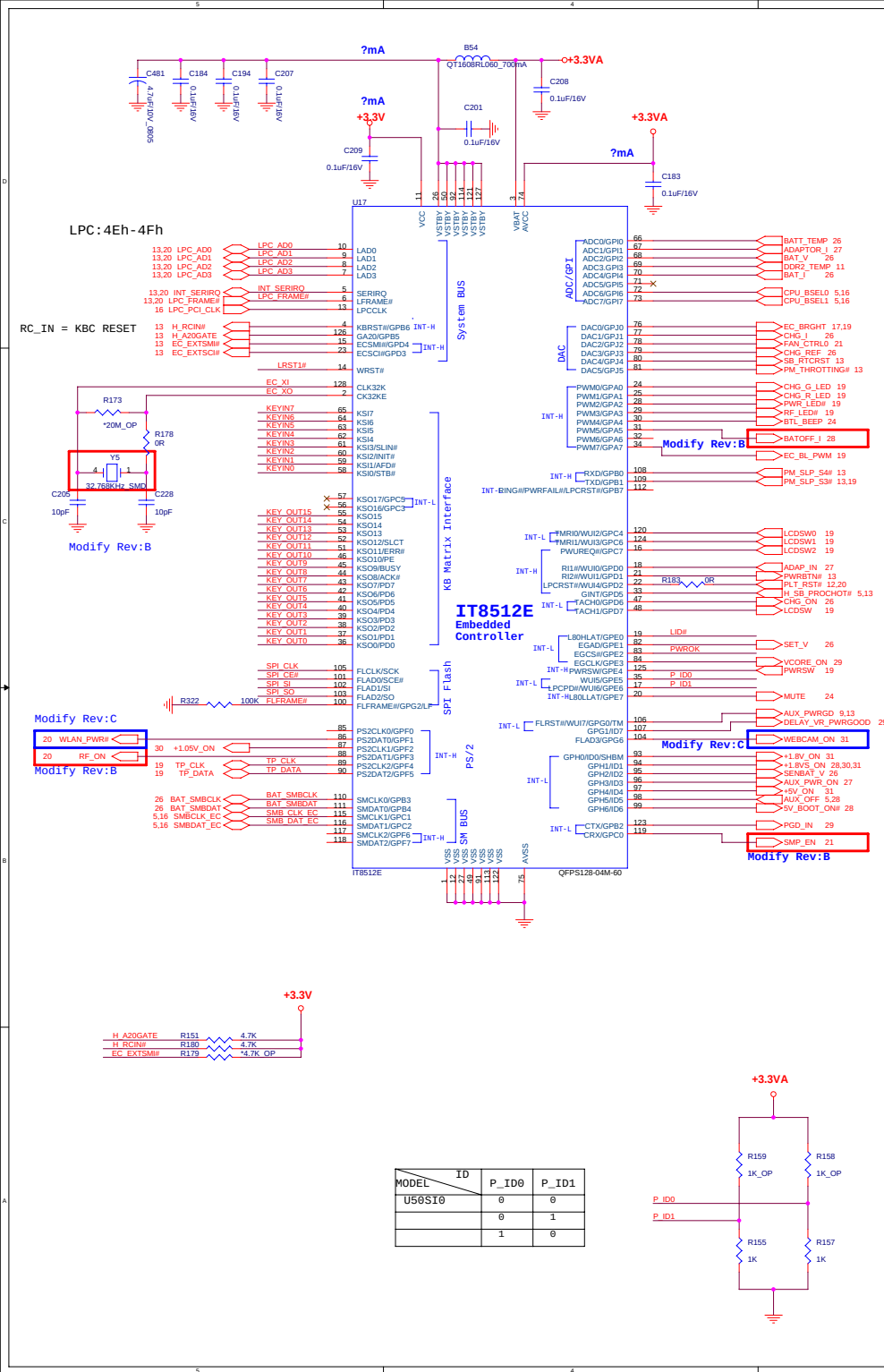




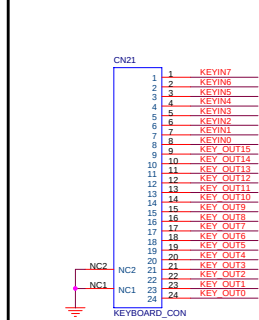


|               |           |        |   |
|---------------|-----------|--------|---|
| HDVBN2 C C118 | 0.1uF/16V | HDVBN2 | 7 |
| HDVPB2 C C120 | 0.1uF/16V | HDVPB2 | 7 |
| HDVBN1 C C114 | 0.1uF/16V | HDVBN1 | 7 |
| HDVPB1 C C116 | 0.1uF/16V | HDVPB1 | 7 |
| HDVBN0 C C110 | 0.1uF/16V | HDVBN0 | 7 |
| HDVPB0 C C112 | 0.1uF/16V | HDVPB0 | 7 |

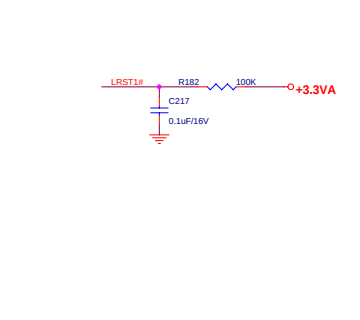
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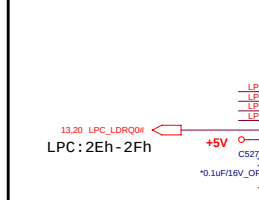
## KEYBOARD



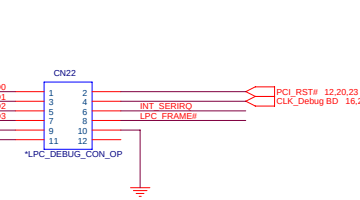
## FLASH ROM(SPI)



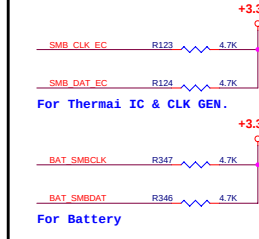
## DEBUG PORT



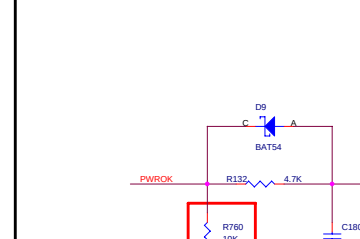
## COVER CLOSE



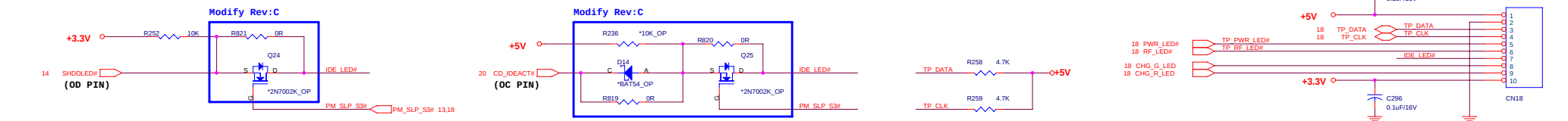
## EC SMBUS



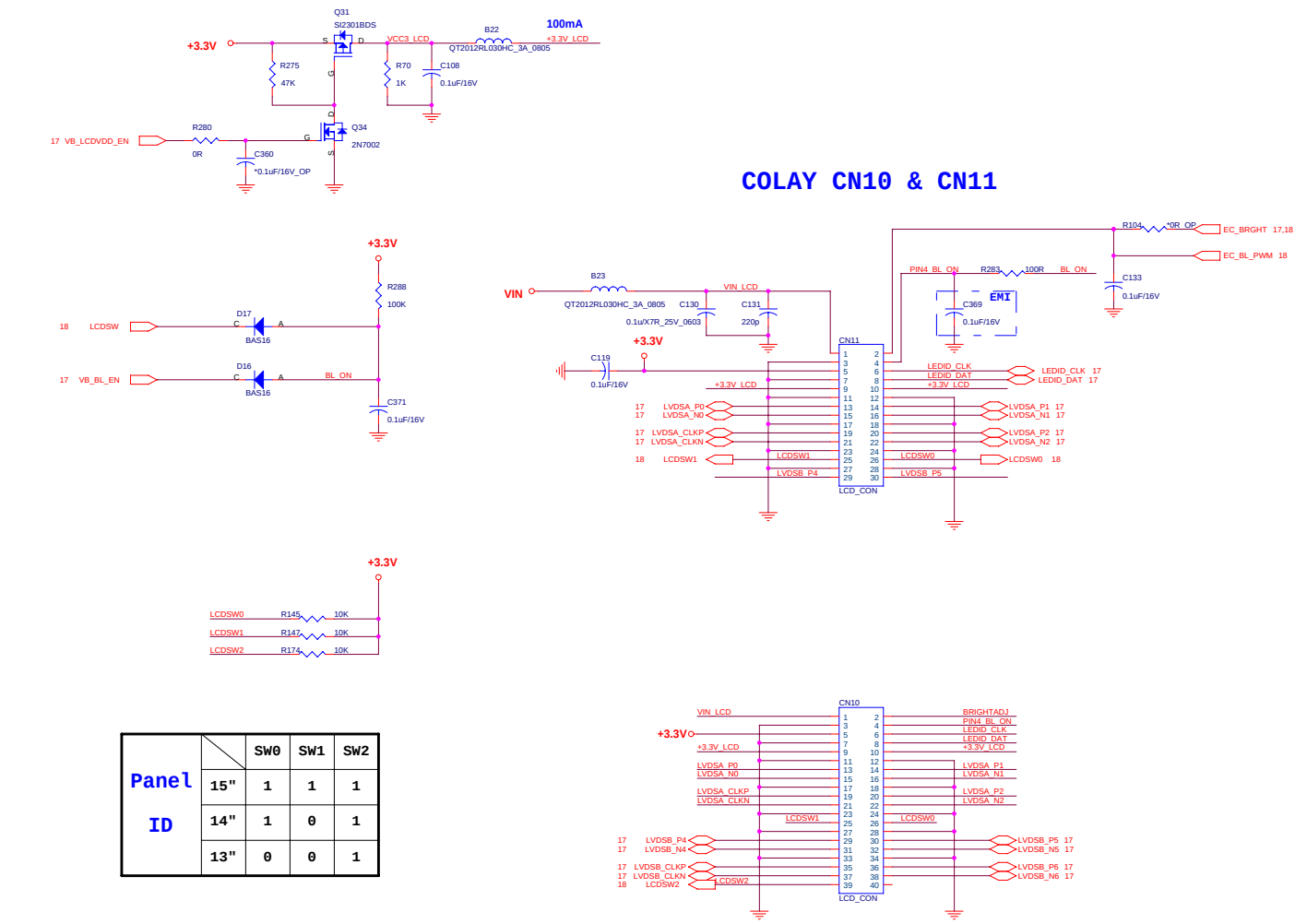
## NB / SB Power OK



Touch Pad

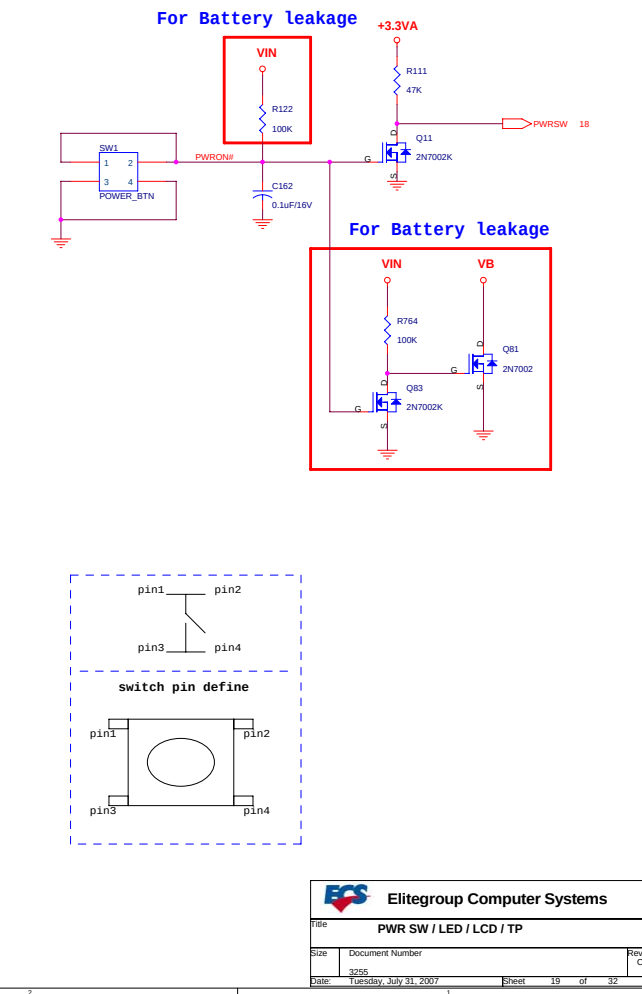


LCD

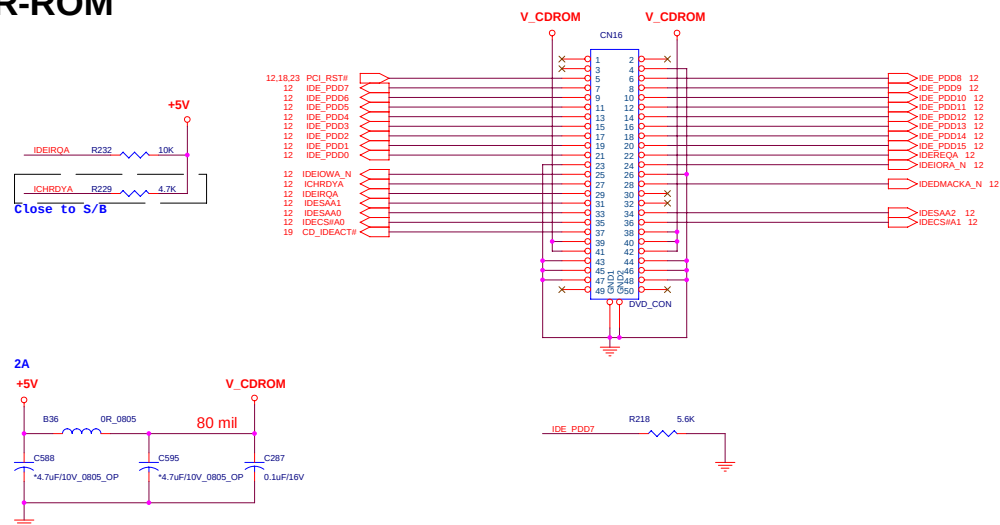


| Panel ID |     | SW0 | SW1 | SW2 |
|----------|-----|-----|-----|-----|
|          | 15" | 1   | 1   | 1   |
|          | 14" | 1   | 0   | 1   |
|          | 13" | 0   | 0   | 1   |

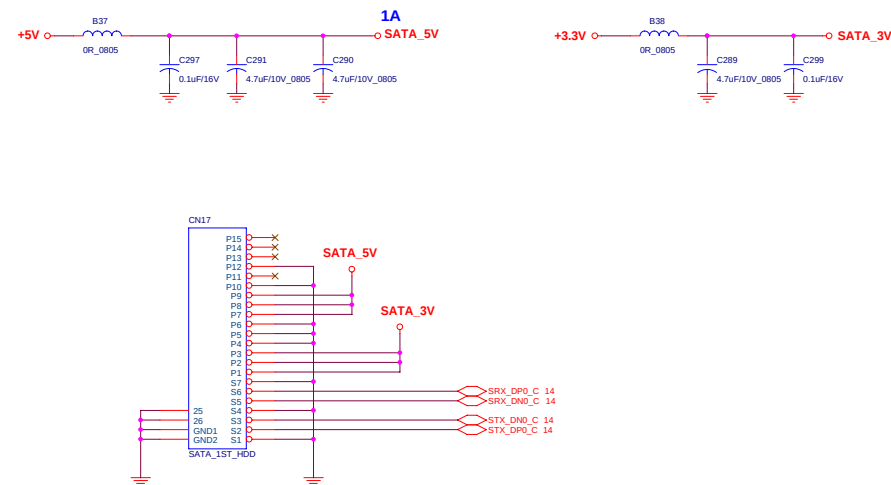
PWR SW



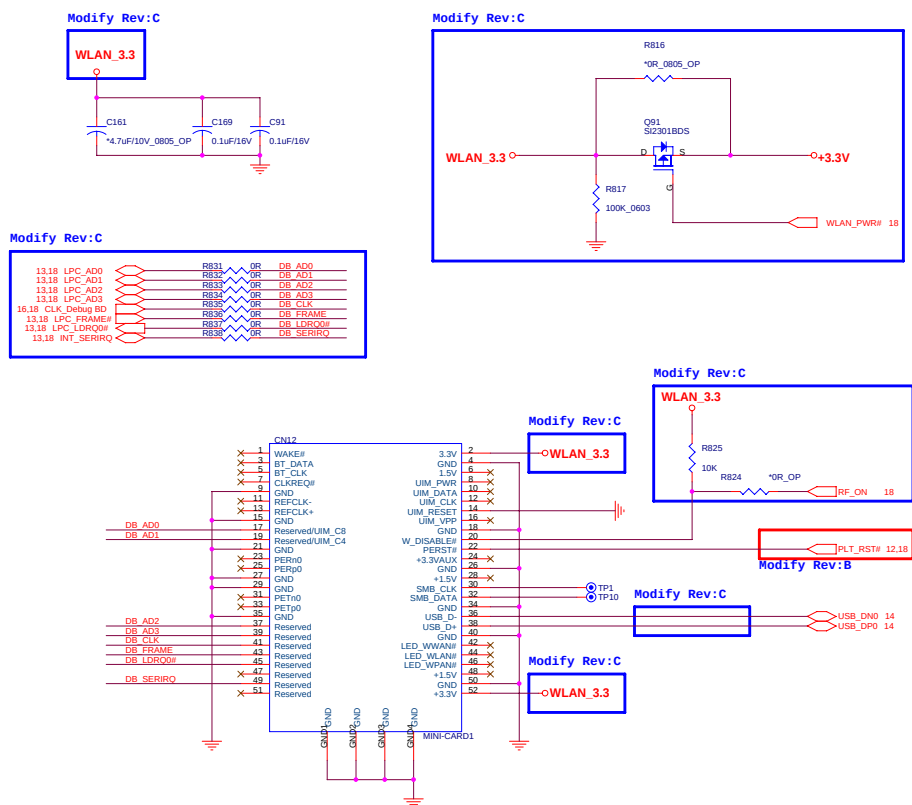
## CR-ROM



## SATA-HDD

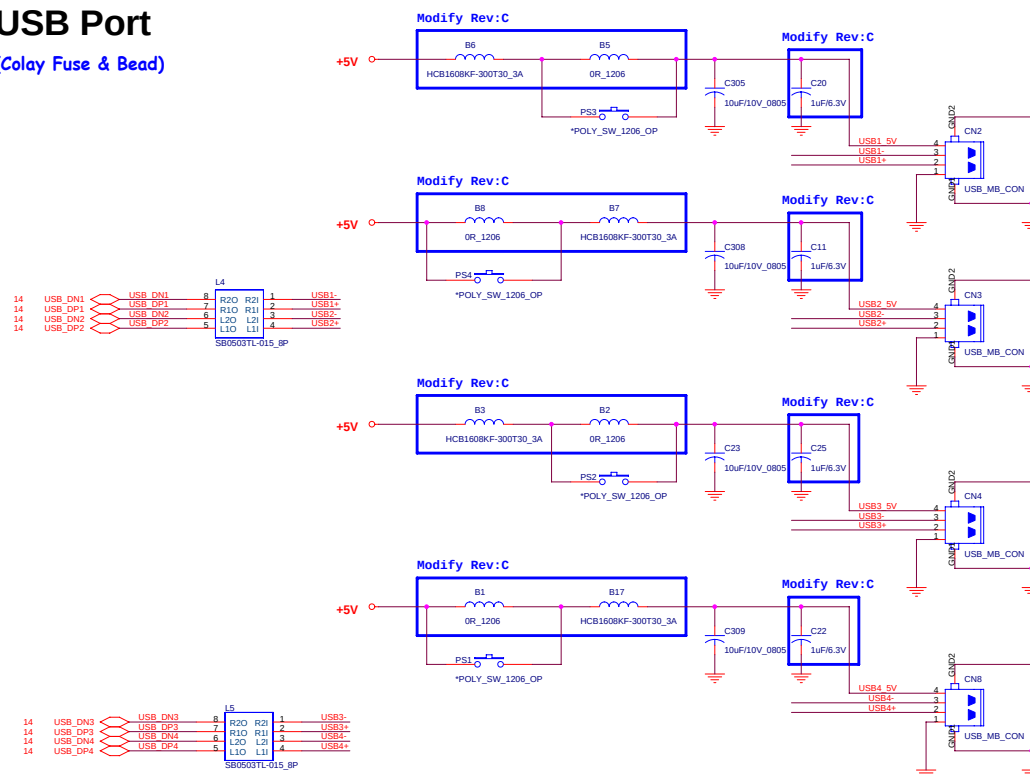


## MINI CARD CONN

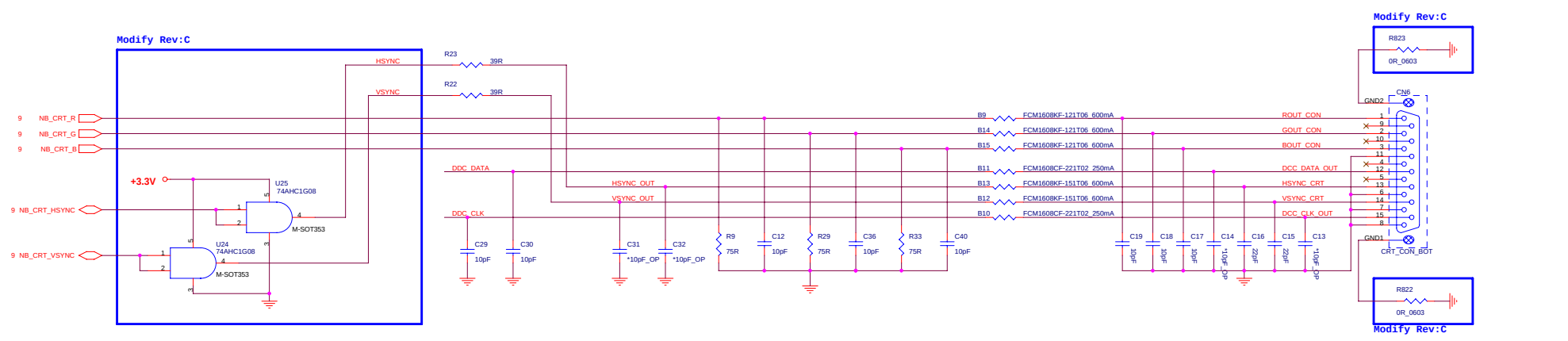


## USB Port

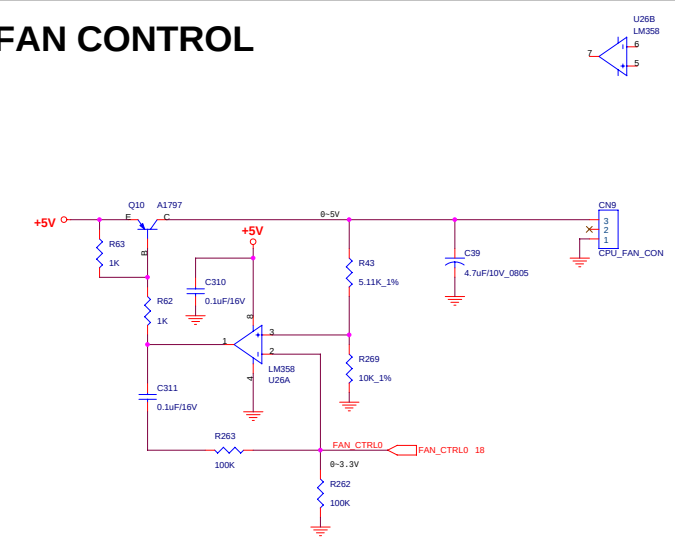
(Colay Fuse & Bead)



CRT

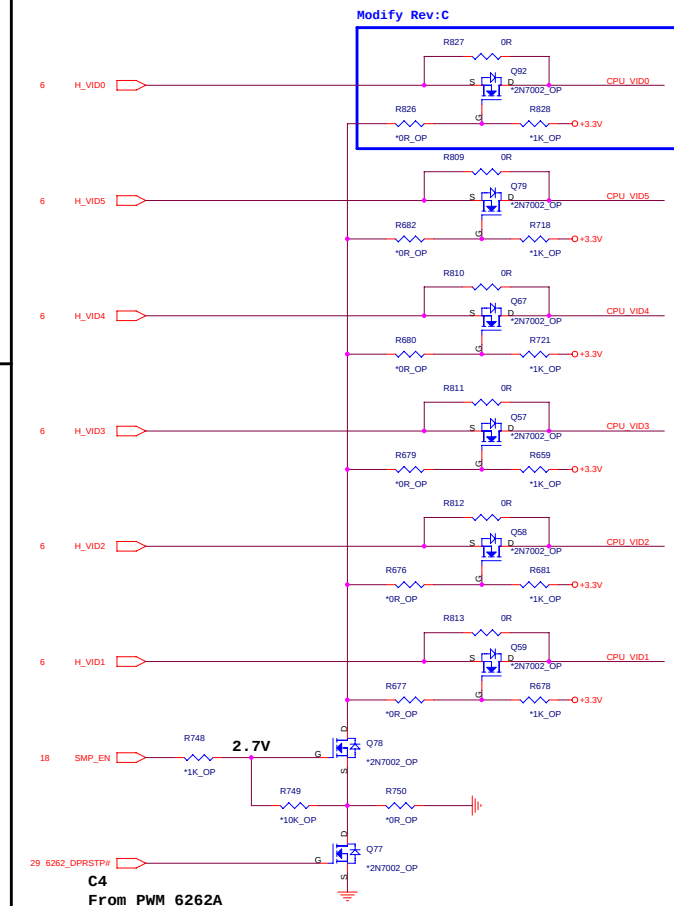


CPU FAN CONTROL

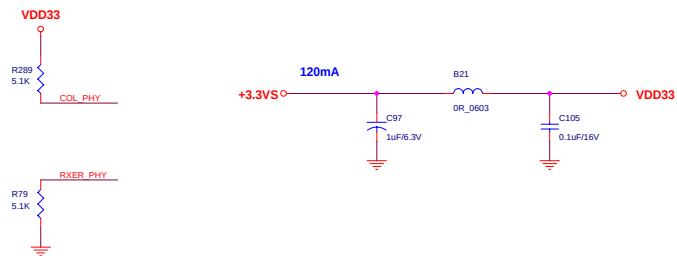


SMART POWER

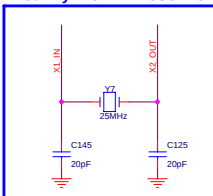
| VID6 | VID5 | VID4 | VID3 | VID2 | VID1 | VID0 | Vcore  | +_mV    |
|------|------|------|------|------|------|------|--------|---------|
| 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1.5000 | -0mV    |
| 0    | 0    | 0    | 0    | 0    | 0    | 1    | 1.4875 | -12.5mV |
| 0    | 0    | 0    | 0    | 0    | 1    | 0    | 1.4750 | -12.5mV |
| 0    | 0    | 0    | 0    | 1    | 0    | 0    | 1.4500 | -25mV   |
| 0    | 0    | 0    | 1    | 0    | 0    | 0    | 1.4000 | -50mV   |
| 0    | 0    | 1    | 0    | 0    | 0    | 0    | 1.3000 | -100mV  |
| 0    | 1    | 0    | 0    | 0    | 0    | 0    | 1.1000 | -200mV  |
| 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0.7000 | -400mV  |



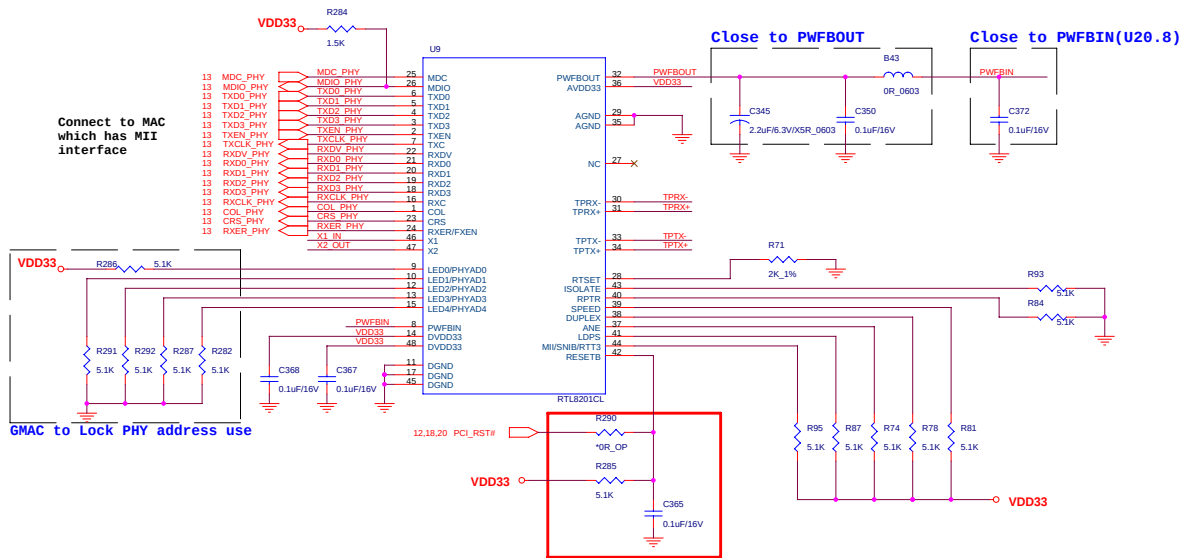




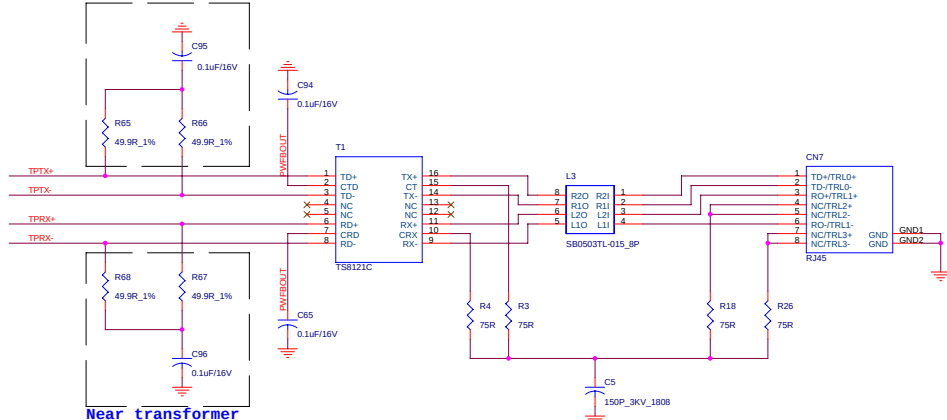
Modify Rev:C  
Modify Rev:B Reserve



Connect to MAC  
which has MII  
interface



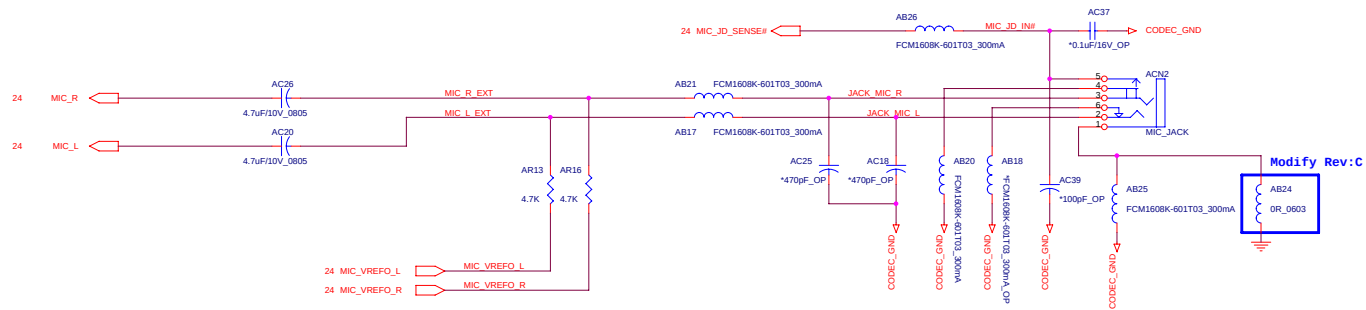
Near LAN PHY



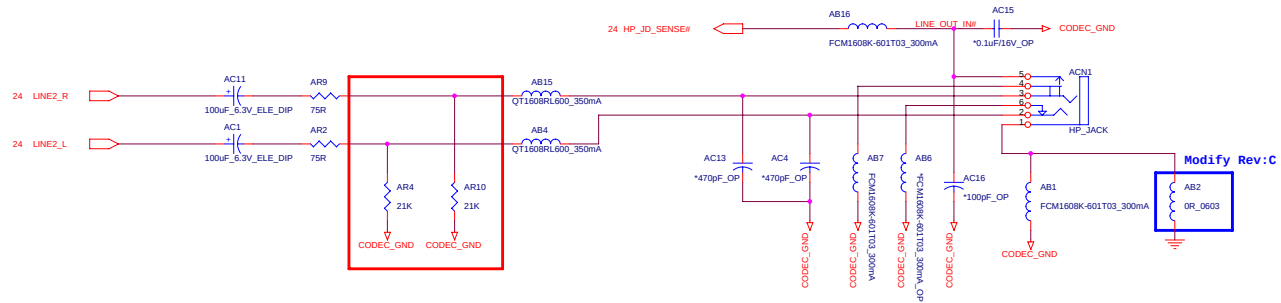




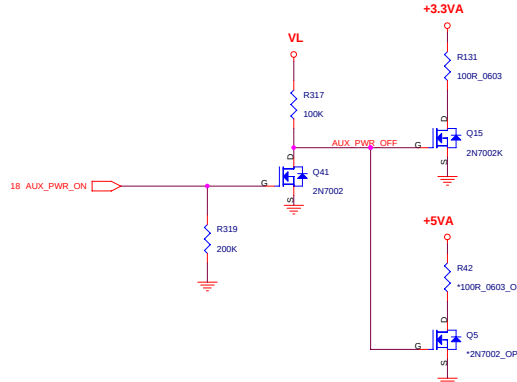
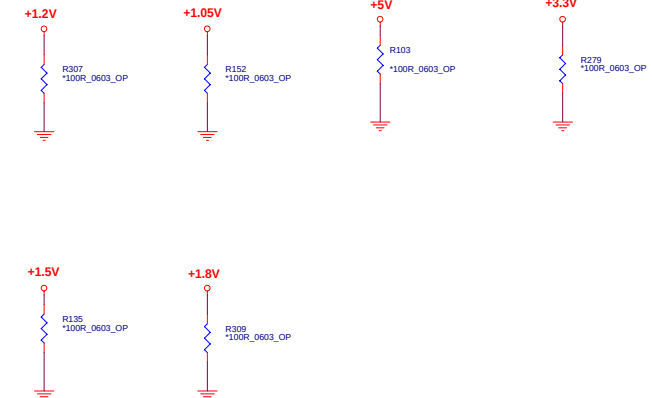
### MIC/Line In JACK



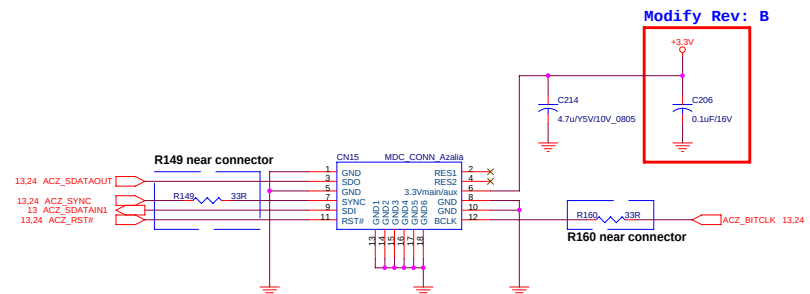
### HeadPhone JACK



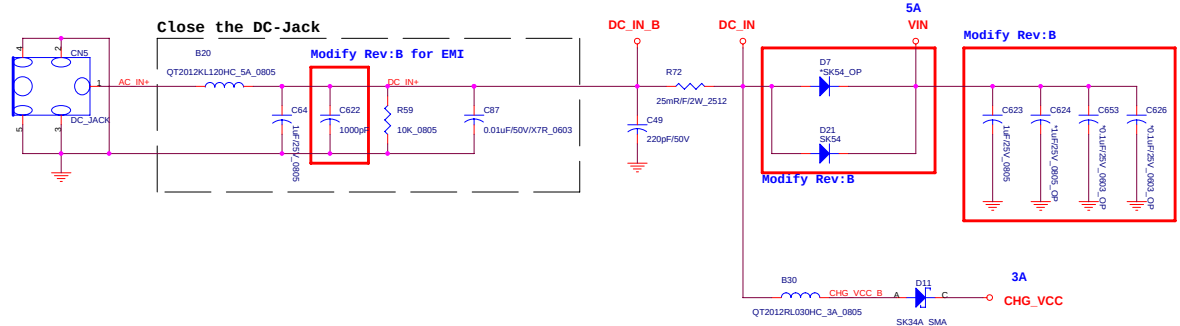




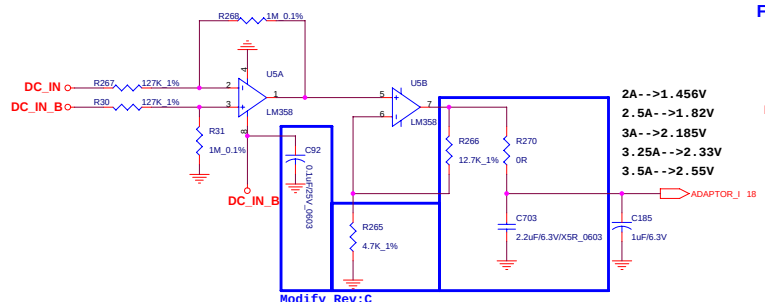
# MDC



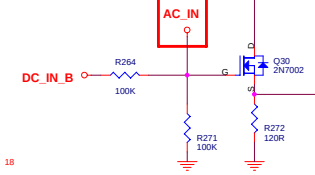
# DC IN



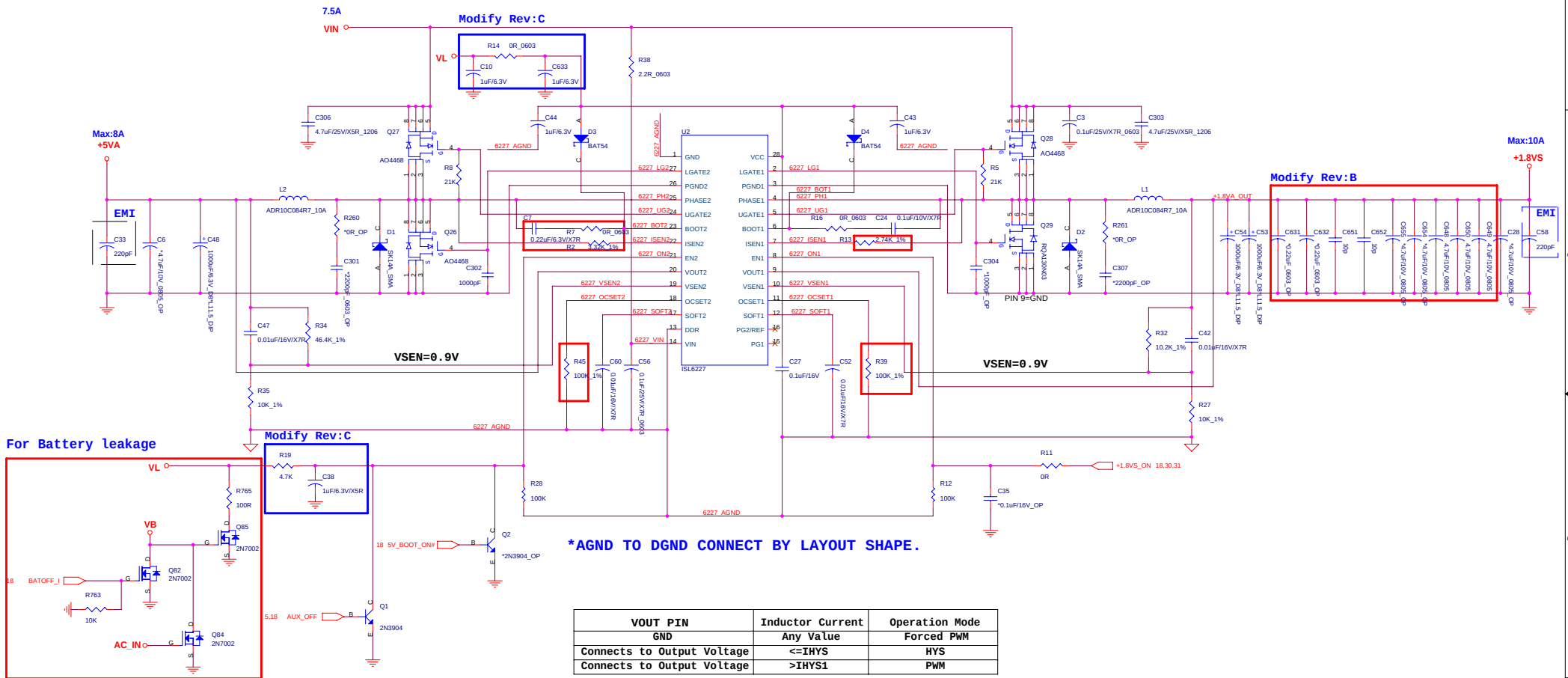
# TOTAL POWER



For Battery leakage



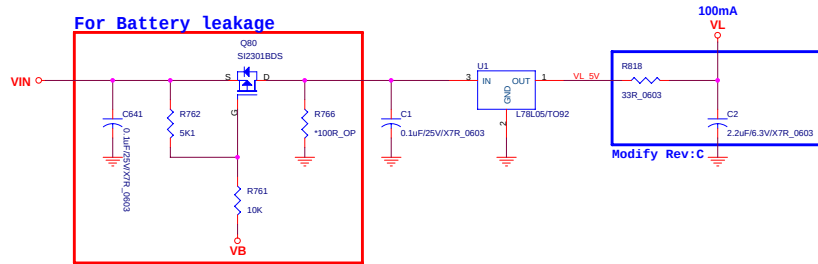
|                             |                        |       |          |
|-----------------------------|------------------------|-------|----------|
| Elitegroup Computer Systems |                        |       |          |
| DC IN/Discharger/APC        |                        |       |          |
| Size                        | Document Number        | Rev C |          |
| 2200                        |                        |       |          |
| Date:                       | Tuesday, July 31, 2007 | Sheet | 27 of 32 |



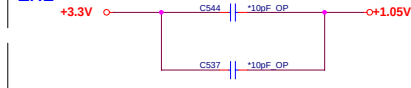


# VL for ISL6227 Boost pin

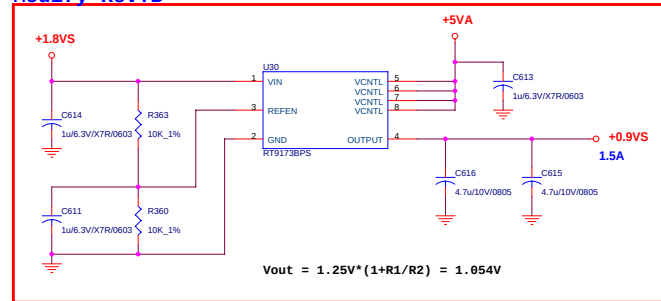
## For Battery leakage



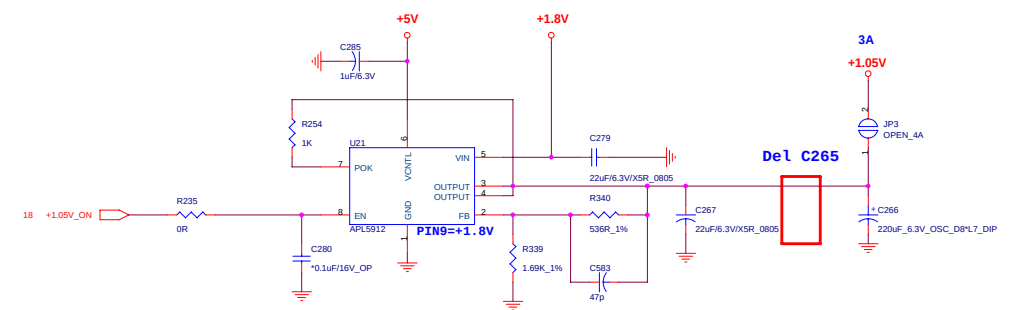
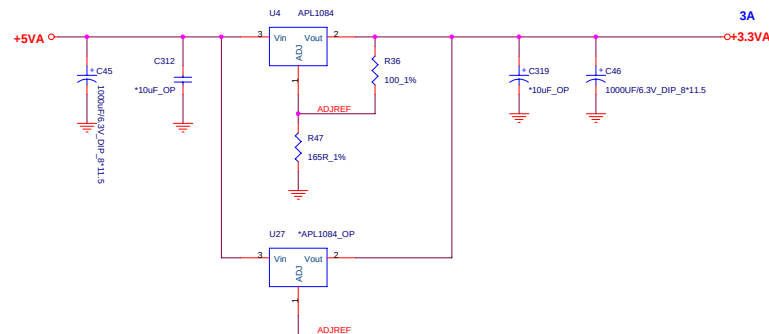
## EMI



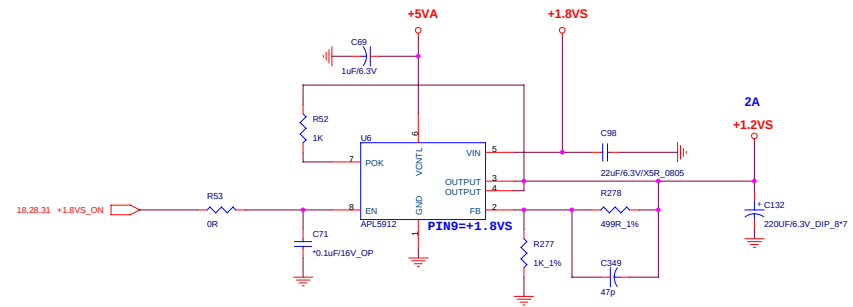
## Modify Rev:B



$$V_{out} = 1.25V \cdot (1 + R1/R2) = 1.054V$$



$$V_{out} = 0.8V \cdot (1 + R1/R2) = 1.054V$$



$$V_{out} = 0.8V \cdot (1 + R1/R2) = 1.2V$$

## Modify Rev:B

